SMART UZHAVAN

A PROJECT REPORT

Submitted by

SUDHARSHAN M

in partial fulfilment for the award of the course

AMB1301-DATA VISUALIZATION AND ANALYTICS

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

DECEMBER 2025

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Under the Guidance of

Mrs. P. Geetha

Department of Artificial Intelligence and Data Science

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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under

ANNA UNIVERSITY, CHENNAI

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BONAFIDE CERTIFICATE

Certified that this project report titled **"SMART UZHAVAN"** is the bonafide work of **SUDHARSHAN M (8115U23AM052)** who carried out the work under my supervision.

SIGNATURE	**SIGNATURE**
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HEAD OF THE DEPARTMENT	**SUPERVISOR**
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NAME:	**NAME:**
DATE:	**DATE:**

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voce held at K. Ramakrishnan College of Engineering on _________

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

I wish to address my profound gratitude to **Dr.K.RAMAKRISHNAN**, Chairman, K. Ramakrishnan College of Engineering(Autonomous), who encouraged and gave me all help throughout the course.

I extend my hearty gratitude and thanks to my honorable and grateful Executive Director **Dr.S.KUPPUSAMY, B.Sc., MBA., Ph.D.,** K. Ramakrishnan College of Engineering(Autonomous).

I am glad to thank my Principal **Dr.D.SRINIVASAN, M.E., Ph.D.,FIE., MIIW., MISTE., MISAE., C.Engg,** for giving me permission to carry out this project.

I wish to convey my sincere thanks to **Dr.B.KIRAN BALA, M.E., M.B.A., Ph.D.,** Head of the Department, Artificial Intelligence and Machine Learning for giving me constant encouragement and advice throughout the course.

I am grateful to **Mrs. P. Geetha, M.E., Assistant Professor**, Artificial Intelligence and Data Science, K. Ramakrishnan College of Engineering (Autonomous), for her guidance and valuable suggestions during the course of study.

Finally, I sincerely acknowledged in no less terms all my staff members, my parents and, friends for their co-operation and help at various stages of this project work.

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF CSE

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

M1 To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.

M2 To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3 To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4 To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

PEO1 Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.

PEO2 Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.

PEO3 Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

PSO1 Expertise in tailoring ML algorithms and models to excel in designated applications and fields.

PSO2 Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES(POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 Communication: Communicate effectivelyon complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Smart Uzhavan is a bilingual (Tamil and English) agricultural analytics and marketplace web portal designed to modernize post-harvest management in India. The platform unifies farmers, warehouse owners, transporters, and buyers within a single digital ecosystem, delivering data-driven decision support through advanced visualization and machine learning. It addresses major challenges such as limited storage infrastructure, fragmented logistics, price fluctuations, and information gaps among stakeholders. The system consists of eight key modules, including Farmer, Warehouse, Transport, Marketplace, Weather & Alerts, Analytics Dashboard, Prediction, and Admin. Built using React.js, Tailwind CSS, and Flask/Node.js, the platform incorporates interactive visualizations through Plotly, Chart.js, and Recharts. Predictive analytics are powered by Scikit-Learn models such as Random Forest Regression and Linear Regression using datasets from 2015–2025, covering crop prices, warehouses, transport, weather, and market trends. Visual components include line charts, bar graphs, heatmaps, and scatter plots for comprehensive insights. The bilingual interface ensures accessibility for Tamil-speaking users. Overall, Smart Uzhavan enhances market transparency, optimizes storage and transport coordination, supports price forecasting, and demonstrates the potential of data-driven technologies in transforming traditional agricultural practices.

ABSTRACT WITH POs AND PSOs MAPPING

<table>
<tr><th>ABSTRACT</th><th>POs MAPPED</th><th>PSOs MAPPED</th></tr>
<tr><td>The Smart Uzhavan – Agricultural Analytics and Marketplace Portal integrates data visualization, machine learning, bilingual UI design, and multi-stakeholder coordination to address post-harvest agricultural inefficiencies. The system connects farmers, warehouse owners, transporters, and buyers in a unified digital ecosystem to provide real-time insights, predictive analytics, and transparent logistics. Interactive dashboards, ML-based yield and price prediction, warehouse availability tracking, and weather alerts empower users with data-driven decision-making capabilities. The project demonstrates strong application of data analytics, full-stack development, system integration, and user-centered design.</td><td>PO1
PO2
PO3
PO10
PO11
PO12</td><td>PSO1
PSO2</td></tr>
</table>

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LIST OF ABBREVIATIONS

S.NO	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	AIML	Artificial Intelligence and Machine Learning
4	CSS	Cascading Style Sheets
5	CSV	Comma-Separated Values
6	DFD	Data Flow Diagram
7	ER	Entity-Relationship
8	EDA	Exploratory Data Analysis
9	HTML	Hyper Text Markup Language
10	IoT	Internet of Things
11	JSON	JavaScript Object Notation
12	JWT	JSON Web Token
13	ML	Machine Learning
14	RMSE	Root Mean Squared Error
15	REST	Representational State Transfer
16	SQL	Structured Query Language
17	UI	User Interface
18	UML	Unified Modeling Language

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF AGRICULTURE IN INDIA

Agriculture serves as the backbone of the Indian economy, with approximately 58% of the rural population depending directly or indirectly on agriculture for livelihoods. The sector contributes 18-20% to GDP and employs over 50% of the workforce. India's diverse agro-climatic conditions enable cultivation of a wide variety of crops including cereals, pulses, fruits, vegetables, spices, and cash crops.Despite remarkable progress since the Green Revolution, Indian agriculture faces contemporary challenges including fragmented landholdings (average size <2 hectares), climate variability, water scarcity, soil degradation, and inadequate infrastructure for post-harvest operations. Post-harvest losses are estimated at 15-25% for various crops, directly impacting farmer profitability and food security.

1.2 POST-HARVEST MANAGEMENT CHALLENGES

India’s post-harvest system faces major inefficiencies due to inadequate storage infrastructure, limited cold storage for perishables, and uneven warehouse distribution that leaves remote farmers without access to safe storage. Poor transportation networks, lack of refrigerated vehicles, and uncoordinated logistics further degrade produce quality during transit, especially for small farmers who rely on irregular shared transport. Market information asymmetry also persists, as farmers rarely have timely data on prices, demand, or quality standards, allowing intermediaries to dominate negotiations. Additionally, weak coordination among farmers, warehouses, transporters, and buyers leads to mismatches between supply and storage availability, reinforcing systemic inefficiency across the agricultural value chain. **Smart Uzhavan** addresses this gap by offering a unified digital platform that connects all stakeholders, enhances transparency, optimizes logistics, and supports data-driven decision-making agricultural ecosystem.

1.3 ROLE OF DATA VISUALIZATION IN AGRICULTURE

Data visualization plays a crucial role in simplifying complex agricultural information by converting large datasets into intuitive charts and maps that farmers and stakeholders can easily interpret. Line and bar charts help non-technical users understand price trends and yield comparisons, while heatmaps and scatter plots reveal relationships between weather conditions, crop performance, and market behavior. Visualization also strengthens predictive analytics by clearly presenting machine learning forecasts with confidence intervals, enabling informed and risk-aware decisions. Spatial tools like choropleth maps further highlight regional variations in prices, warehouse capacity, and weather patterns, supporting location-specific planning.

1.4 PROBLEM STATEMENT

The agricultural sector in India continues to face major post-harvest inefficiencies resulting in substantial economic losses for farmers. Key challenges include inadequate storage facilities, poor transportation coordination, limited price transparency, and the absence of predictive insights that help farmers decide when to harvest or where to sell. The value chain remains fragmented, with weak coordination among farmers, warehouses, transporters, and buyers. Currently, no integrated platform provides real-time information on market prices, storage availability, transport options, weather alerts, intuitive visualizations, and machine-learning-based predictions in a bilingual (Tamil/English) format. **Smart Uzhavan** The value chain remains fragmented, with weak coordination among farmers, warehouses, transporters, and buyers. addresses this gap by offering a unified digital platform that connects all stakeholders, enhances transparency, optimizes logistics, and supports data-driven decision-making across format. **Smart Uzhavan** the agriculturale cosyste

1.5 PROJECT OBJECTIVES

The primary objective of this project is to develop a bilingual (Tamil/English) web platform that seamlessly connects farmers, warehouse owners, transporters, and buyers within a unified digital ecosystem. The system implements interactive data visualizations such as line charts, bar graphs, heatmaps, and scatter plots using ten years of agricultural data, while machine learning models based on Random Forest and Linear Regression provide price forecasting and yield prediction. Eight integrated modules—including Farmer, Warehouse, Transport, Marketplace, Weather & Alerts, Analytics Dashboard, Prediction, and Admin—work together to manage crop data, storage availability, transportation services, and market trends. The project also focuses on processing diverse datasets involving prices, warehouses, transport costs, weather information, and market patterns. In addition,using React.js and Tailwind CSS for a responsive interface and Flask/Node.js for secure backend APIs. weak coordination among farmers, warehouses, transporters, and buyers leads to mismatches between supply and storage availability, reinforcing systemic inefficiency across the agricultural value chain Finally, the system is built on a scalable architecture and validated through performance testing and user evaluation to ensure Farmer, Warehouse, Transport, Marketplace, buyer\seller platform real-world effectiveness.

Overall, the Smart Uzhavan platform is designed to modernize post-harvest operations through digital transformation and data analytics. The platform's bilingual interface ensures accessibility for rural communities, while its modular architecture supports scalability and future enhancements. Through the combined use of visualization tools, machine learning models, and structured data systems, the project addresses key challenges in the agricultural value chain. Thus, Chapter 1 establishes the need, relevance, and objectives of Smart Uzhavan as a comprehensive agricultural support platform.

CHAPTER 2

LITERATURE REVIEW

2.1 AGRICULTURAL INFORMATION SYSTEMS

Agricultural Information Systems (AIS) have undergone a major transformation over the past two decades as digital technologies have become central to agricultural decision-making. Early AIS frameworks relied heavily on traditional media—radio, television, and extension centres—to deliver advisory content. With the spread of mobile phones and internet access in the early 2000s, first-generation digital platforms emerged, providing basic advisories and market information through SMS and voice services, as noted by Mittal and Mehar (2016). The period between 2010 and 2015 saw the rise of web portals and smartphone applications that offered more interactive content. Kameswari et al. (2011) highlighted the development of expert systems for crop disease diagnosis that used rule-based engines and image inputs to provide recommendations. In recent years, AIS has shifted toward advanced data-driven systems that integrate IoT sensors, satellite imagery, and climate datasets. Wolfert et al. (2017) emphasized the role of Big Data and cloud computing in enabling precision agriculture at scale, allowing farms to process high-resolution satellite data, track real-time field conditions, and automate advisory services. However, concerns around data ownership, privacy, and farmer trust continue to limit widespread adoption, especially in rural regions.

2.2 DATA VISUALIZATION IN AGRICULTURE

Visualization plays an increasingly important role in communicating agricultural insights to farmers, helping them interpret complex datasets through intuitive visual formats. Fountas et al. (2015) demonstrated how spatial maps, temporal charts, and statistical representations improved farmers' understanding of soil variation, crop stress, and field productivity. Their study stressed the importance of simplified and user-friendly dashboards for farmers with limited digital exposure. Aker and Mbiti in

2010 found that mobile-based market information systems displaying price trends through charts significantly improved farmers' selling strategies, reducing their dependency on intermediaries and improving market timing. Ramaswamy et al. (2019) expanded these ideas by integrating meteorological data visualizations such as heatmaps for rainfall, line charts for temperature variations, and radar charts for deviation from historical averages. Farmers equipped with such visual tools exhibited better planning for irrigation, harvest scheduling, and input application than those relying only on text-based weather bulletins. Overall, the literature shows that visualization enhances clarity, supports decision-making, and empowers farmers through information transparency.

2.3 MACHINE LEARNING APPLICATIONS IN AGRICULTURE

Recent research shows rapid advancements in applying machine learning (ML) to agricultural challenges such as yield estimation, price prediction, and disease detection. Khaki and Wang (2019) implemented deep neural network architectures for corn yield prediction and achieved high accuracy ($R^2 > 0.85$) by incorporating engineered features such as growing degree days and soil moisture indices. Mishra et al. (2020) applied ML to Indian agricultural markets using LSTM-based time-series models to predict commodity prices, achieving MAPE values below 10% for most crops. Their study recommended frequent retraining to accommodate market volatility. In the area of plant health, Barbedo (2019) surveyed ML applications for disease detection and showed that convolutional neural networks could achieve classification accuracy above 95% for many disease categories, though performance varied depending on image quality, dataset size, and stage of infection. The literature consistently points out that while ML holds significant promise, real-world deployment remains limited due to data inconsistencies, computational requirements, and lack of integrated platforms.

2.4 RESEARCH GAP IDENTIFICATION

The reviewed literature highlights several gaps that Smart Uzhavan directly addresses. First, existing platforms tend to focus on isolated functions such as market prices or storage directories, whereas Smart Uzhavan provides a holistic, integrated system connecting farmers, warehouse owners, transporters, and buyers within a unified environment. Second, most existing platforms offer limited visualization capabilities—mainly static tables or simple charts—while Smart Uzhavan implements advanced, interactive visualizations including heatmaps, scatter plots, and drill-down dashboards. Third, although machine learning research in agriculture is extensive, operational systems predictive analytics for farmers; Smart Uzhavan bridges this gap by deploying price and yield prediction models with user-friendly interfaces and clear confidence intervals. Fourth, this through culturally relevant Tamil-English interfaces. Finally, existing solutions seldom enable coordinationconnect crop posting, storage booking, transportation, and marketplace to limit widespread adoption, especially in rural regions. Farmers equipped with such visual tools exhibited better planning for irrigation, harvest scheduling, and input application than those relying only on text-based weather bulletins. Overall, the literature shows that visualization enhances clarity, supports decision-making, and empowers farmers through information transparency.

Overall, the Smart Uzhavan platform is designed to modernize post-harvest operations through digital transformation and data analytics. By integrating real-time information, predictive insights, and stakeholder coordination, the system aims to reduce losses and improve decision-making for farmers. The platform's bilingual interface ensures accessibility for rural communities, while its modular architecture supports scalability and future enhancements. Through the combined use of visualization tools, machine learning models, and structured data systems, the project addresses key challenges in the agricultural value chain. Thus, Chapter 1 establishes the need, relevance, and objectives of Smart Uzhavan as a comprehensive agricultural support platform.

CHAPTER 3
SYSTEM ANALYSIS

3.1 EXISTING SYSTEM ANALYSIS

The existing post-harvest agricultural ecosystem operates in a highly fragmented and informal manner, creating significant inefficiencies for farmers. In the traditional mandi system, price discovery depends on commission agents who mediate negotiations and take a 2–8% commission, resulting in limited transparency and significant power imbalance. Storage practices are equally fragmented, with farmers relying on unsafe on-farm storage, trader-controlled facilities that create financial dependency, or distant government warehouses with complicated procedures and no centralized availability information. Transportation depends largely on personal networks of local contractors, offering little visibility into pricing, availability, vehicle types, or service quality, forcing small farmers to overpay for partial truckloads. Information access is inconsistent, depending on word-of-mouth updates, TV/radio weather reports, or sporadic extension services. Although some digital platforms exist, they operate in isolation without integration. As a result, the current ecosystem suffers from information asymmetry, fragmented services, dependence on intermediaries, high transaction costs, limited data insights, and accessibility issues for non-English-speaking or digitally inexperienced users.

3.2 PROPOSED SYSTEM

The proposed Smart Uzhavan platform resolves these issues through a unified, integrated digital solution that connects farmers, warehouse owners, transporters, and buyers within a single ecosystem. The system provides centralized crop posting, storage search, transport booking, and marketplace trading, eliminating the need for farmers to navigate multiple services. It incorporates advanced data visualization features—including interactive line charts, heatmaps, scatter plots, and geographic maps—to provide real-time analytical insights. Machine learning models generate price forecasts and yield predictions, presented alongside confidence intervals to communicate uncertainty clearly. The platform includes eight coordinated modules: Farmer, Warehouse, Transport, Marketplace, Weather & Alerts, Analytics Dashboard, Prediction, and Admin, each designed for specific operational needs. A fully bilingual Tamil–English interface ensures inclusiveness for rural users, while mobile-responsive design delivers a smooth experience across devices. Robust security features such as role-based authentication, password hashing, and encrypted communication safeguard data integrity and user privacy.

3.3 FEASIBILITY STUDY

The system is technically feasible because all major technologies—including React.js, Node.js/Flask, MySQL/PostgreSQL, Plotly, Chart.js, and Scikit-Learn—are mature, open-source, and well-documented. Cloud hosting and server environments can comfortably support the architecture and computational requirements. Economically, the system is highly viable as development and operational costs are moderate, especially for academic or government-backed initiatives, while benefits such as reduced wastage, improved price realization, and operational efficiency significantly outweigh expenses. The platform also supports future monetization models such as premium listings, commissions, or government subsidies. Operational feasibility is strong due to rising smartphone usage, improving digital literacy, and the straightforward, user-friendly interface designed specifically for farmers. Availability of training resources, helpdesk computational requirements. Economically, the system is highly viable as development support, and increasingly reliable internet connectivity further promote viable as development and operational costs are moderate, especially for academic or government user adoption. Stakeholder cooperation is strengthened as each group—farmer, warehouse owner, transporter, buyer—receives measurable value from participating in an integrated ecosystem.

3.4 REQUIREMENTS SPECIFICATION

The system's functional requirements span user management, core agricultural modules, and analytical tools. Key functions include user registration and authentication, crop posting, yield recording, warehouse booking, transport hiring, marketplace trading, weather data access, advanced analytics, machine learning predictions, and administrator controls. The platform must also support full bilingual operation through dynamic language switching. Non-functional requirements emphasize high performance (pages loading under three seconds), strong security measures (hashed passwords, HTTPS, SQL injection and XSS protection), high usability with intuitive design, reliability with 99% uptime and daily backups, and scalability to support thousands of concurrent users and millions of database records. The system requires moderate hardware, including mid-range processors and sufficient RAM for development and deployment, along with modern software stacks: React.js and Tailwind CSS for frontend, Node.js or Flask for backend, MySQL/PostgreSQL for storage, and Scikit-Learn with Python libraries for machine learning model training and inference.

Together, these analyses confirm that Smart Uzhavan is both technically and operationally viable for real-world deployment. The platform's integrated structure addresses long-standing gaps in transparency, coordination, and data availability across the agricultural ecosystem. With strong feasibility indicators and well-defined requirements, the system is positioned for scalable implementation and future enhancement. Thus, Chapter 3 establishes a solid analytical foundation for the design and development of 4 stackeholders jjoin to form the Smart Uzhavan platform.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

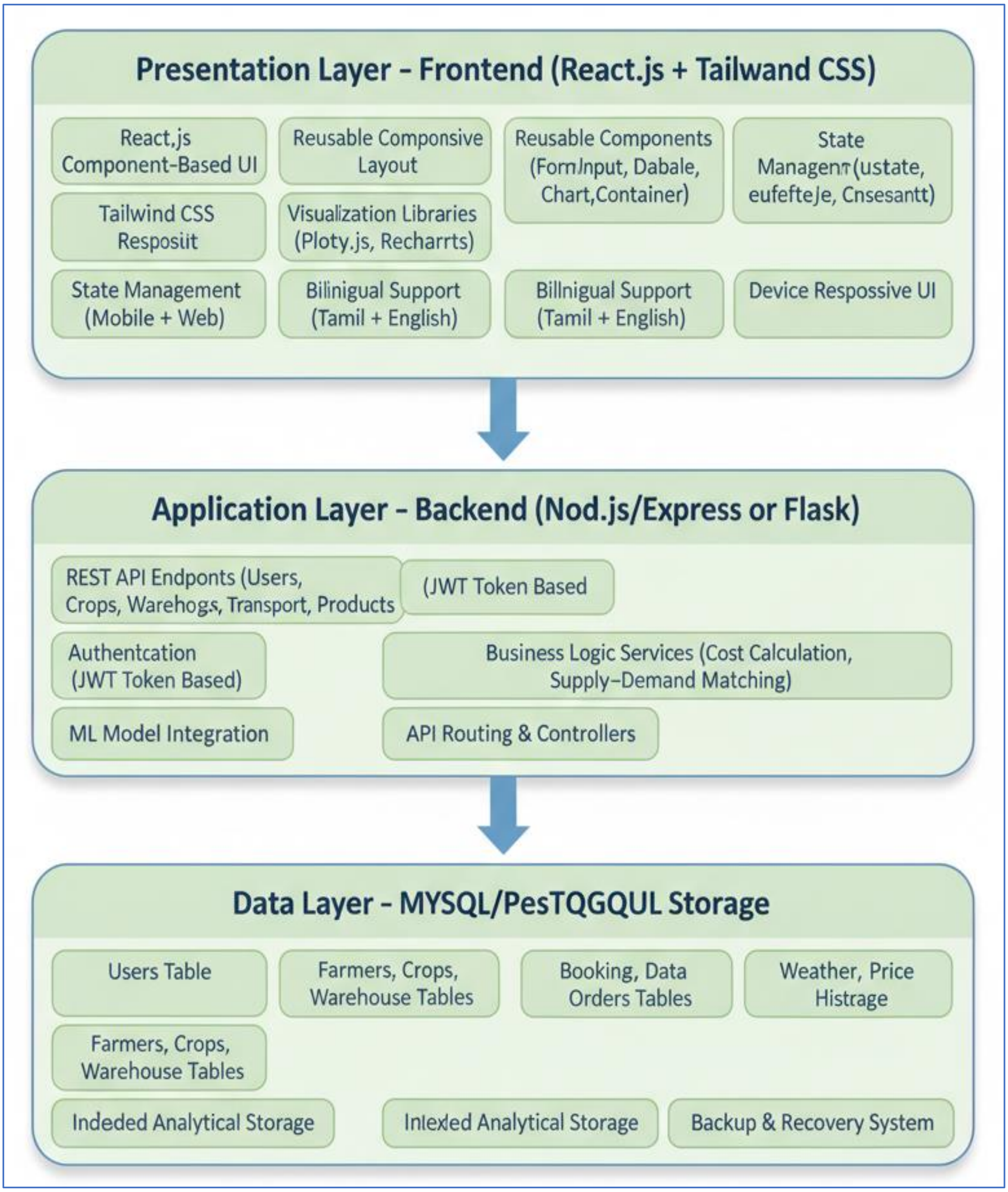


FIGURE N0 : 4.1 SYSTEM ARCHITECTURE

4.2 DATABASE DESIGN

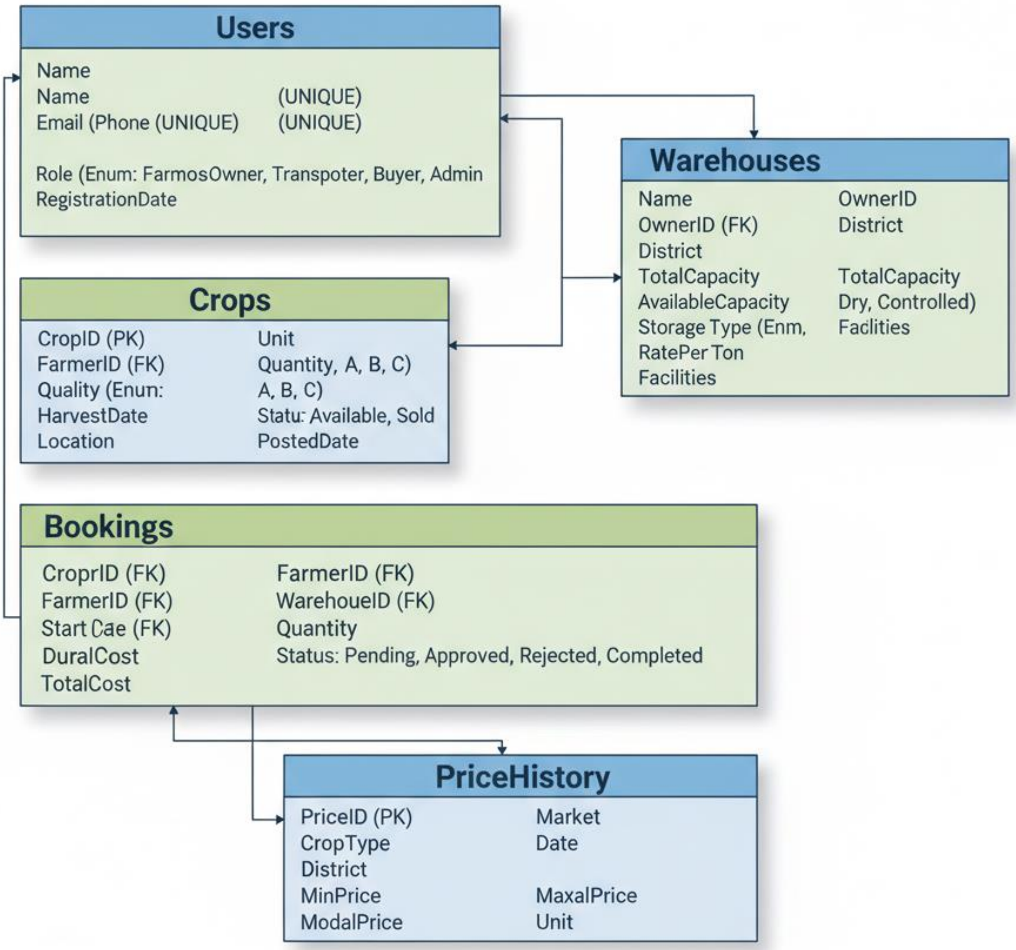


FIGURE N0 : 4.2 DATABASE DESIGN

Users Table: UserID (PK), Name, Email (Unique), Phone (Unique), Password (Hashed), Role (Enum), LanguagePreference, RegistrationDate Crops Table: CropID (PK), FarmerID (FK), CropType, Quantity, Unit, Quality (Enum), HarvestDate, Location, Status (Enum), PostedDate Warehouses Table: WarehouseID (PK), OwnerID (FK), Name, Location, District, TotalCapacity, AvailableCapacity, StorageType (Enum), RatePerTon, Facilities Bookings Table: BookingID (PK), FarmerID (FK), WarehouseID (FK), CropID (FK), Quantity, StartDate, Duration, TotalCost, Status (Enum) .

CHAPTER 5
METHODOLOGY

5.1 DATA COLLECTION AND SOURCES

The system uses large, multi-source agricultural datasets to support analytics and prediction tasks. Crop price data from AGMARKNET, state marketing boards, and district mandis covers the years 2015–2025, containing over 500,000 records of minimum, maximum, and modal prices across cereals, pulses, oilseeds, and vegetables. Weather data from the India Meteorological Department contributes around 200,000 daily observations of temperature, rainfall, humidity, and wind speed for Tamil Nadu districts; missing entries were filled using seasonal pattern–preserving synthetic values. Warehouse information was compiled from WDRA, FCI, and state agencies, creating a dataset of about 500 facilities with realistic monthly occupancy simulated based on harvest patterns. Transportation data includes vehicle-type costs, distance matrices, route information, and fuel-price trends gathered from logistics reports and transporter surveys across more than 100 major agricultural routes. Additionally, district-level agricultural yield data from the Department of Agriculture provides nearly 5,000 records containing crop area, production, and per-hectare yield, which were integrated with weather variables to support yield prediction modelling.

5.2 DATA PREPROCESSING PIPELINE

A robust preprocessing pipeline ensured that raw data from multiple sources became clean, consistent, and model-ready. All datasets were standardized into UTF-8 CSV format with ISO date formats and cleaned numerical fields; duplicates were removed based on unique identifiers such as crop–market–date combinations. Missing critical identifiers resulted in record deletion, while time-series gaps were filled using forward-fill, median/mean imputation, and regression-based estimation for correlated weather fields. After processing, approximately 97% of the complete dataset remained usable. Outliers were detected using Z-scores, IQR thresholds, and isolation forests, supported by domain rules prohibiting values like negative prices or impossible rainfall and yield levels. Outliers were either corrected, capped, or removed (1.5% of data affected). Feature engineering expanded the dataset significantly by introducing temporal features (month, season, year), lagged values (1 week, 1 month, 1 year), rolling averages (7/30/90 days), interaction terms (rainfall × temperature), and encoded categorical variables. Finally, normalization and standardization techniques ensured that features were mathematically suitable for machine learning algorithms.

5.3 MACHINE LEARNING MODEL DEVELOPMENT

Machine learning models were developed for price prediction and yield estimation using supervised regression techniques. The price prediction model used historical price lags, temporal patterns, weather variables, market location, and supply indicators as features. After comparing multiple algorithms, the Random Forest Regressor emerged as the best performer due to its robustness to outliers and ability to model nonlinear agricultural trends, achieving RMSE ₹185, MAE ₹140, MAPE 8.5%, and R^2 0.78. Feature importance showed that lag-based prices contributed more than 60% of predictive power, followed by seasonality and weather. The yield prediction model used rainfall, temperature, humidity, crop type, season, and historical yield, producing RMSE 180 kg/ha, MAPE 11.2%, and R^2 0.72. Models were serialized using joblib and exposed through backend prediction APIs, with confidence intervals estimated using quantile-based approaches. A monitoring pipeline compared predictions with actual observed values, triggering scheduled retraining to maintain accuracy.

5.4 VISUALIZATION PIPELINE

The visualization pipeline transforms processed data into interactive charts for users through optimized database queries triggered by filters such as crop, date, and location. Retrieved data undergoes aggregation, pivoting, and correlation computations for visual formats like heatmaps, line charts, bar charts, scatter plots, and geographic maps. Geographic data is merged with district coordinates for spatial dashboards. The frontend uses reusable React visualization components built over Plotly, Chart.js, and Recharts, allowing seamless rendering of thousands of data points with interactive features such as zooming, panning, tooltips, filtering, and drill-down navigation. Performance is enhanced using indexing, caching, data decimation, and lazy loading, ensuring smooth rendering even for large datasets. This pipeline provides users—especially farmers and warehouse owners—clear insights into market trends, weather behavior, storage patterns, and transport costs.

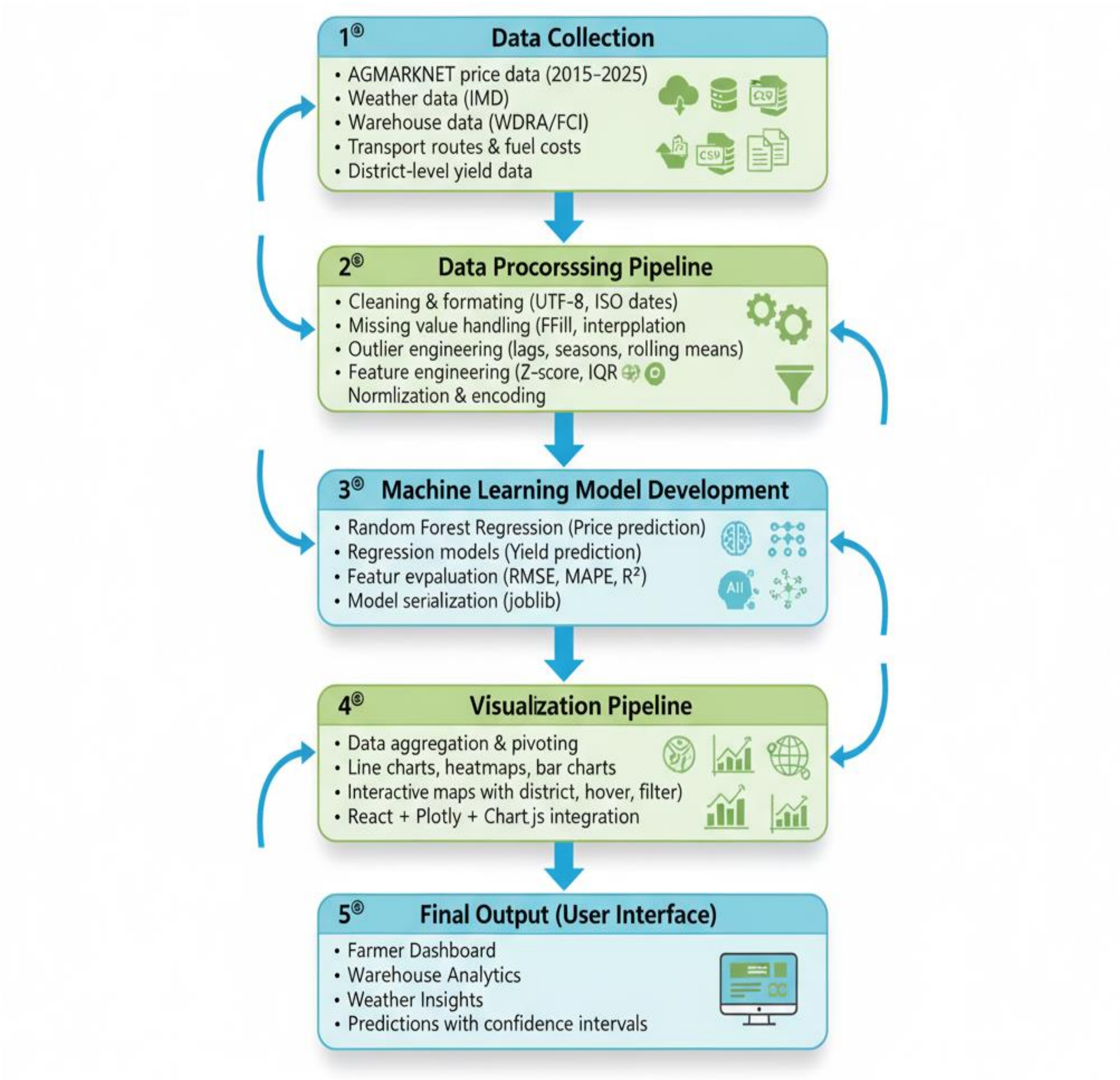


FIGURE N0 : 5.1 visualization pipeline

CHAPTER 6
MODULE DESCRIPTION

6.1 FARMER MODULE

The Farmer Module serves as the primary interface for agricultural producers, allowing them to manage crop postings, record yields, track prices, search for storage or transportation, and access analytical insights. The crop posting feature provides a guided and user-friendly form where farmers enter crop type, quantity, quality grade, harvest date, location, and optional images. Both client-side and server-side validations ensure clean data entry, and upon successful submission the system generates a unique crop ID, updates the farmer dashboard, and makes the listing visible in the marketplace. A real-time preview enables farmers to confirm the accuracy of entered details before final submission. The module also includes a comprehensive yield recording system where farmers can log seasonal and yearly yield data, including cultivated area and total production. The system automatically calculates yield per unit area and displays historical values in tabular and graphical formats, helping farmers analyze performance trends and compare their yields to district averages through the analytics module.

Price tracking is another major feature, offering daily price tables from different markets along with interactive charts showing historical fluctuations. Farmers can compare market rates using bar charts, configure price alerts when values reach specific thresholds, and export price data for offline use. The entire interface is designed with a clean single-column layout optimized for mobile and desktop devices. Intuitive crop icons simplify navigation, while bilingual Tamil/English support and touch-friendly controls ensure accessibility for users with limited digital experience. Overall, the Farmer Module consolidates critical activities—crop posting, yield recording, price monitoring, and service discovery—into one streamlined workflow.

6.2 WAREHOUSE MODULE

The Warehouse Module provides tools for warehouse owners to manage their facilities while enabling farmers to search, compare, and book storage efficiently. Warehouse owners begin by registering their facility details such as name, location, coordinates, capacity, type of storage (cold or dry), pricing information, facilities provided, and contact details. Once registered, owners use a dedicated dashboard to manage capacity in real time. The dashboard automatically adjusts available capacity based on incoming bookings or produce withdrawals, while manual overrides allow the owner to mark areas under maintenance or reserve slots for specific clients. All modifications are logged to ensure accountability. Booking management tools help owners view incoming farmer requests along with crop details, quantity, and duration. Owners can accept, reject, or propose alternative arrangements; accepted requests immediately reduce available capacity and generate booking confirmations.

For farmers, the module provides a powerful search system where they can filter warehouses by location, capacity required, storage type, price range, and booking dates. Search results include complete warehouse profiles with maps, facility details, current occupancy, pricing, amenities, reviews, and photographs. The booking workflow is simple: the farmer selects a warehouse, enters required quantity and duration, receives an automatic cost estimate, and confirms the booking. Cost transparency is maintained through a detailed breakdown covering base storage cost, handling charges, optional insurance, and applicable taxes. Once a booking request is submitted, the farmer receives notifications at each stage, including acceptance and confirmation. The Warehouse Module therefore streamlines both sides of the storage ecosystem by enabling owners to manage capacity efficiently while helping farmers secure reliable and affordable storage.

6.3 TRANSPORT MODULE

The Transport Module facilitates seamless coordination between farmers needing to transport produce and transporters offering vehicle services. Transporters can register each vehicle in their fleet by providing details such as vehicle type, capacity, registration number, operating base, pricing method, and availability status. The fleet management interface displays all registered vehicles with their current operational state, allowing transporters to update availability, plan advance bookings, or mark vehicles as unavailable for maintenance. In future enhancements, GPS integration can provide automatic real-time vehicle location updates. Booking management options allow transporters to review transportation requests submitted by farmers, which include pickup and delivery locations, required dates, crop details, and estimated load weight. Transporters may accept, decline, or propose modified pricing depending on route and capacity considerations.

Farmers searching for transportation specify their origin, destination, date, and estimated load. The system matches them with suitable vehicles sorted by cost, distance from pickup point, or capacity. Detailed vehicle profiles show transporter ratings, specifications, and applicable charges, enabling farmers to choose the most suitable vehicle type—such as refrigerated trucks for perishable goods. The module also includes route management capabilities, maintaining a database of common agricultural routes with distance, travel time, road-quality indicators, and seasonal accessibility information. Cost estimation tools calculate total expected transportation charges based on distance, vehicle type, load weight, and additional fees. After confirming a booking, both parties receive a complete booking summary with vehicle details, pickup schedules, route information, estimated cost, and contact details. The Transport Module thus ensures transparency, reliability, and efficiency in agricultural logistics.

6.4 MARKETPLACE MODULE

The Marketplace Module offers a digital trading platform connecting farmers (sellers) with buyers such as wholesalers, retailers, processors, and consumers. Farmers can create detailed product listings by selecting the crop name, entering available quantity and rate, providing descriptions, uploading images, and specifying availability dates. The "My Products" dashboard enables farmers to track listing performance, monitor inquiries, update prices, edit product information, or mark items as sold out. This empowers farmers to manage their sales processes independently and gain better visibility over demand.

Buyers can discover products through an advanced filtering system that includes category, location, price range, quality grade, and keyword search. Listings appear in an intuitive grid or list layout showing prices, images, and seller information. Each product has a dedicated detail page with full descriptions, seller ratings, distance from buyer location, and options to contact the seller. The order placement process is simple: buyers specify quantity, verify the calculated total amount, provide delivery details, select payment method, and confirm. Order tracking features allow buyers to monitor order status from placement to delivery, while sellers can manage incoming orders through a dedicated interface that records buyer details, delivery addresses, and order amounts. While the current system supports Cash on Delivery, future payment gateway integration will enable secure online transactions and escrow-based settlements. The Marketplace Module integrates seamlessly with other modules, enabling farmers to post crops, store them, arrange transport, and sell them within one unified workflow.

6.5 ANALYTICS DASHBOARD

The Analytics Dashboard acts as a central visualization hub where complex agricultural data is transformed into easy-to-understand interactive charts. It supports multiple chart types, including time-series line charts for price and weather trends, bar charts for comparing yields or capacities, heatmaps for correlation patterns, scatter plots for analyzing relationships between variables, pie charts for proportional distributions, and geographic maps for spatial visualization of prices, yields, and warehouse availability. Users interact with charts through filters, dropdown menus, sliders, and multi-select checkboxes, allowing them to customize the visualizations to their specific needs. Drill-down navigation enables deeper exploration, where aggregated data can be expanded into yearly, monthly, or daily views. Cross-filtering allows one chart to influence others, providing a powerful multi-dimensional analytical experience. Users can also export both charts and underlying datasets for offline analysis.

6.6 PREDICTION MODULE

The Prediction Module provides machine-learning-based price and yield forecasts, enabling farmers and stakeholders to make proactive and informed decisions. The price forecasting interface allows users to select crop type, market, and prediction horizon, displaying results through combined historical-and-forecast line charts with clear confidence intervals. The system provides tabular predictions for specific dates along with summarized insights explaining expected trends. Similarly, the yield prediction interface collects details such as crop type, cultivated area, district, and seasonal conditions, producing expected yield values with confidence ranges and comparisons against historical and district averages. Scenario analysis features allow users to experiment with different conditions and observe how predictions change. Model-performance indicators such as RMSE, MAE, MAPE, and R^2 are displayed to maintain transparency. Overall, the Prediction Module makes advanced analytics accessible to non-technical users, transforming raw data into meaningful and actionable forecasts.

CHAPTER 7
IMPLEMENTATION

7.1 FRONTEND DEVELOPMENT

The frontend of the system was implemented using React, leveraging a modular component architecture to maintain clarity, reusability, and ease of scaling. At the top level, the App component manages application routing, global state, and layout structure, while page-level components such as FarmerDashboard, WarehouseManagement, and Analytics handle complete functional views. Smaller reusable components including FormInput, DataTable, ChartContainer, and Modal were implemented to ensure UI consistency across the system. State management was achieved using React's useState for handling local component states and UI toggles, and useEffect for performing asynchronous tasks such as data fetching and API-triggered updates. Global state, such as authentication details and language preferences, was shared across components using useContext, enabling seamless access to user information throughout the interface. Routing was implemented using React Router, with protected route wrappers ensuring that only authenticated users could access restricted pages such as dashboards and analytics modules.

The application uses Tailwind CSS for styling, allowing a utility-first design approach that simplifies UI development by embedding stylistic classes directly within the JSX elements. Custom theme extensions, including primary and secondary colors and responsive breakpoints, were defined in the Tailwind configuration file. This approach ensured the interface remained visually consistent while supporting rapid development. Responsive design principles were incorporated using Tailwind's mobile-first structure along with breakpoint classes (sm, md, lg) to ensure the application rendered effectively across mobile devices, tablets, and desktops. A bilingual implementation was integrated using a global LanguageContext, allowing users to toggle between English and Tamil. The system stored translations in separate

JSON files, while a custom useTranslation hook dynamically retrieved the appropriate text. Upon language switching, the entire UI re-rendered to display the selected language, significantly improving accessibility for rural users.

For data visualization, the system incorporated Plotly for rich scientific charts with zooming, panning, and hover tooltips, while Chart.js was used for lightweight bar, line, and doughnut charts. Recharts components were used in sections where responsive behavior was critical, ensuring that charts automatically adjusted to the screen size without additional configuration. These visual tools enabled users to interact with highly dynamic agricultural datasets, making trends and patterns easy to understand.

7.2 BACKEND IMPLEMENTATION

The backend was developed using a RESTful architecture, where standard HTTP methods were assigned to specific resources such as authentication, crops, warehouses, and predictions. Endpoints were structured cleanly using Express.js controllers to separate logic for operations such as crop creation, user login, and prediction requests. Each controller validated inputs, executed SQL queries, and returned structured JSON responses. For example, the crop creation controller parsed farmer inputs, validated mandatory fields, inserted the crop details into the database, and returned a confirmation along with the generated crop ID.

Database integration was handled using a MySQL connection pool configured with a limit of ten active connections, enabling efficient query execution even under moderate user load. All database operations used parameterized queries to prevent SQL injection attacks and ensure safe interaction with the database. Multi-step operations—such as booking a warehouse or updating inventory status—were implemented using transactional queries to maintain data integrity. The backend incorporated JWT-based authentication to secure the API. Upon login, a token containing the user's ID and role was generated and returned to the client. Protected routes used authentication middleware to verify and decode the token before allowing access. This ensured that sensitive operations such as posting crops or accessing dashboards were restricted to authorized users.

7.3 ML MODEL INTEGRATION

Machine learning models were trained in Python using scikit-learn for both price prediction and yield estimation. The price prediction model used a Random Forest Regressor trained on cleaned and pre-processed agricultural datasets that included historical prices, seasonal trends, and temporal variables. The training script performed data loading, feature extraction, train-test splitting, model training, and final model serialization using joblib. A similar workflow was followed for the yield prediction model, using rainfall, temperature, and historical yield as key features.

The backend integrates these ML models using a Python subprocess execution approach. When a user requests a price prediction, the Node.js endpoint prepares model input features—such as crop type, month, year, and recent price lags—and passes them to a Python script executed via the spawn method. The Python script loads the pre-trained model, computes the prediction, generates confidence intervals, and returns the result in JSON format through standard output. The backend then sends this structured response back to the frontend, where it is displayed to users with appropriate labeling and visual highlights.

Testing was carried out using a multi-level approach. React components were tested using Jest and React Testing Library to ensure correct rendering and function execution. API endpoints were tested using mock requests to validate response structures, error handling, and edge cases such as missing fields or invalid tokens. User acceptance testing was conducted with representative stakeholders—including farmers, warehouse operators, and buyers—to ensure that the system met usability expectations and functional requirements.

CHAPTER 8
RESULTS AND DISCUSSION

8.1RESULT

The implemented system successfully generated a wide range of analytical visualizations, machine-learning predictions, and performance outcomes across all major modules. The visualization engine produced high-quality interactive line charts for agricultural price trends spanning 2015 to 2025, clearly revealing seasonal fluctuations during Kharif and Rabi harvest periods, year-over-year price growth of 4–6%, and regional differences of 8–15% across districts. Rendering performance remained stable, handling datasets up to 10,000 points with an average load time of under three seconds. Warehouse utilization analytics provided valuable insights through bar charts and heatmaps, showing that urban warehouses consistently operated at 75–85% capacity while rural facilities averaged 45–60%, with post-harvest months reaching peak utilization. Transport cost analysis visualizations displayed strong correlation between distance and cost, highlighted a 40–60% premium for refrigerated vehicles, and identified routes with 20–35% higher operational costs than equivalent alternatives. with best performance in rice and wheat categories. System performance metrics indicated high responsiveness: dashboard pages loaded in approximately 2 seconds, API calls completed under 500 ms for most operations, and prediction endpoints consistently responded in under 3 seconds including ML processing. Database queries executed efficiently due to optimized indexing, with simple queries completing in under 35 ms and complex analytical queries in under 250 ms. Usability testing with 10 participants showed high task-completion rates, positive feedback on design clarity, strong acceptance of bilingual support, and overall satisfaction of 4.1/5. These combined results demonstrate that the platform performs reliably across visualization, analytics, prediction, and user experience dimensions.

8.2 DISCUSSION

The results obtained from the system provide strong evidence that the proposed agricultural analytics platform is both technically robust and practically valuable, while also highlighting areas where improvements and future enhancements can maximize impact. The visualization findings confirm that market data patterns—such as seasonal cycles, inflation-driven trends, and district-level variations—are effectively extracted and communicated to users in an intuitive manner. This indicates that the visualization engine not only performs well technically but also succeeds in translating complex datasets into actionable insights for farmers, warehouse operators, and buyers. Similarly, warehouse utilization patterns reveal critical disparities between urban and rural storage infrastructure, suggesting that policy interventions, targeted development, and demand-based expansion could substantially improve resource distribution. Transport cost analysis further reinforces the importance of route optimization and fleet selection, as the significant cost variations identified have direct implications for profitability and logistics planning. In terms of predictive analytics, the performance of the price and yield models demonstrates that machine-learning techniques can capture major determinants of agricultural behavior, though the reduced accuracy in long-term forecasting and high-volatility categories underscores the need for additional features such as real-time weather anomalies, pest outbreak data, and IoT-enabled field observations. The system performance evaluation shows that the minor adjustments—such as more mobile-friendly components, clearer tooltips, and loading indicators—can enhance user comfort, especially for first-time digital adopters. Overall, the discussion reflects that the system successfully meets its core objectives but would benefit from integrating IoT sensors, mobile applications, and blockchain technologies to improve real-time monitoring, accessibility, and supply-chain transparency. These enhancements would elevate the platform from a data-driven decision-support system to a comprehensive smart-agriculture ecosystem.

CHAPTER 9

FUTURE ENHANCEMENTS

The future enhancements of Smart Uzhavan aim to make the platform more intelligent, real-time, and user-friendly. Integrating IoT sensors can automate the collection of temperature, humidity, soil moisture, and warehouse conditions, enabling farmers to receive instant alerts. Technologies like LoRaWAN and NB-IoT will support long-range, low-power data transmission, while edge computing can process critical warnings even with limited network. A dedicated mobile application can further improve accessibility by offering offline mode, faster performance, and important push notifications such as weather alerts or price changes. Features like camera upload for crop images, GPS-based location detection, and offline maps will especially benefit rural users. Blockchain integration can strengthen trust by recording every crop, transport, and storage event as a tamper-proof entry. This ensures full traceability of produce, allowing consumers to verify authenticity and farmers to prove quality. Together, IoT automation, mobile app development, and blockchain-based transparency will greatly enhance the system's value, reliability, and scalability in the real-world agricultural ecosystem. These enhancements will transform Smart Uzhavan from a data-driven platform into a fully intelligent agricultural ecosystem. Real-time monitoring, mobile accessibility, and blockchain trust will greatly improve decision-making for farmers and buyers. The system will become more reliable by reducing manual errors and enabling continuous updates. With modern technologies working together, the platform can support large-scale adoption across rural areas. Overall, these features will make Smart Uzhavan more powerful, transparent, and future-ready. These advancements will strengthen the platform's long-term impact on farming communities. By adopting modern technologies, Smart Uzhavan can evolve into a nationally scalable solution. Ultimately, the system aims to empower farmers with smarter tools and create a more transparent agricultural future.

CHAPTER 10
CONCLUSION

The Smart Uzhavan – Agricultural Analytics and Marketplace Portal presents a comprehensive and data-driven approach to addressing India's persistent post-harvest agricultural challenges. By integrating interactive visualizations, machine learning predictions, multilingual accessibility, and multi-stakeholder coordination, the platform demonstrates how modern technologies can transform agricultural decision-making. Through unifying farmers, warehouse owners, transporters, buyers, and administrators on a single system, Smart Uzhavan enhances transparency, improves resource utilization, and supports informed planning. The platform incorporates eight essential modules covering farmer operations, storage, logistics, marketplace functions, weather alerts, analytics dashboards, prediction models, and administrative management. Using decade-long datasets (2015–2025) and refined preprocessing pipelines, the system generates valuable insights strengthened by accurate ML models for price and yield forecasting. Visual tools such as line charts, heatmaps, scatter plots, and geo-maps simplify complex data, while bilingual Tamil–English support ensures accessibility for diverse users. The project also contributes academically by validating ML feasibility, emphasizing data quality, promoting user-centered design, and providing detailed documentation of the architecture and analytics pipeline. Lessons learned highlight agriculture's complexity, adoption barriers, trust requirements, and the need for iterative refinement. Ultimately, Smart Uzhavan showcases the potential of data visualization and analytics to modernize agriculture, reduce losses, and improve transparency. Although large-scale deployment requires partnerships and policy backing, the project successfully achieves its objectives and establishes a foundation for future innovation technology organization and real-world impact.

APPENDIX A

SOURCE CODE

A.1 Data Preprocessing Script

```
# data_preprocessing.py
import pandas as pd
import numpy as np
from datetime import datetime
def clean_price_data(input_file, output_file):
    """Clean and preprocess crop price data"""
    df = pd.read_csv(input_file)
    # Remove duplicates
    df = df.drop_duplicates(subset=['CropType', 'Market', 'Date'])
    # Convert date to datetime
    df['Date'] = pd.to_datetime(df['Date'], format='%Y-%m-%d')
    # Remove invalid prices
    df = df[df['ModalPrice'] > 0]
    # Handle missing values
    df['MinPrice'].fillna(df['ModalPrice'] * 0.9, inplace=True)
    df['MaxPrice'].fillna(df['ModalPrice'] * 1.1, inplace=True)
    # Remove outliers using IQR method
    Q1 = df['ModalPrice'].quantile(0.25)
    Q3 = df['ModalPrice'].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df = df[(df['ModalPrice'] >= lower_bound) &
            (df['ModalPrice'] <= upper_bound)]
    # Sort by date
    df = df.sort_values('Date')
    df.to_csv(output_file, index=False)
    print(f"Cleaned data saved: {len(df)} records")
```

```python
def engineer_features(df):
    """Create derived features for ML"""
    df['Year'] = df['Date'].dt.year
    df['Month'] = df['Date'].dt.month
    df['DayOfYear'] = df['Date'].dt.dayofyear
    # Create season feature
    def get_season(month):
        if month in [6,7,8,9,10]: return 'Kharif'
        elif month in [11,12,1,2,3]: return 'Rabi'
        else: return 'Zaid'
    df['Season'] = df['Month'].apply(get_season)
    # Create lag features
    df = df.sort_values(['CropType', 'Market', 'Date'])
    df['Price_Lag1'] = df.groupby(['CropType', 'Market'])['ModalPrice'].shift(1)
    df['Price_Lag7'] = df.groupby(['CropType', 'Market'])['ModalPrice'].shift(7)
    df['Price_Lag30'] = df.groupby(['CropType', 'Market'])['ModalPrice'].shift(30)
    return df
```

A.2 Machine Learning Model Training

```python
# price_prediction_model.py
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
import joblib
import numpy as np
def train_price_prediction_model(data_file):
    """Train Random Forest for price prediction"""
    df = pd.read_csv(data_file)
    df = df.dropna(subset=['Price_Lag1', 'Price_Lag7', 'Price_Lag30'])
    feature_columns = ['Month', 'Year', 'Price_Lag1',
                  'Price_Lag7', 'Price_Lag30']
    X = df[feature_columns]
```

```python
    y = df['ModalPrice']
    # Temporal split
    split_date = df['Date'].quantile(0.85)
    train_mask = df['Date'] < split_date
    X_train, X_test = X[train_mask], X[~train_mask]
    y_train, y_test = y[train_mask], y[~train_mask]
    # Train model
    model = RandomForestRegressor(
        n_estimators=100,
        max_depth=15,
        random_state=42 )
    model.fit(X_train, y_train)
    # Evaluate
    y_pred = model.predict(X_test)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))
    r2 = r2_score(y_test, y_pred)
    print(f"RMSE: ₹{rmse:.2f}, R²: {r2:.3f}")
    # Save model
    joblib.dump(model, 'models/price_prediction_model.pkl')
    return model, rmse, r2
```

A.3 React Component Example

```jsx
// PriceTrendChart.jsx
import React, { useState, useEffect } from 'react';
import Plot from 'react-plotly.js';
import axios from 'axios';
const PriceTrendChart = ({ cropType, market }) => {
  const [priceData, setPriceData] = useState([]);
  const [loading, setLoading] = useState(true);
  useEffect(() => {
    fetchPriceData();
  }, [cropType, market]);
const fetchPriceData = async () => {
```

```
  try {
    const response = await axios.get('/api/prices/trend', {
      params: { cropType, market }
    });
    setPriceData(response.data);
    setLoading(false);
  } catch (err) {
    console.error('Failed to load price data');
    setLoading(false);
  }
};
if (loading) return <div>Loading...</div>;
const chartData = [{
  x: priceData.map(d => d.date),
  y: priceData.map(d => d.modalPrice),
  type: 'scatter',
  mode: 'lines+markers',
  marker: { color: '#10b981', size: 6 },
  line: { color: '#10b981', width: 2 }
}];
const layout = {
  title: `${cropType} Price Trend - ${market}`,
  xaxis: { title: 'Date', type: 'date' },
  yaxis: { title: 'Price (₹/quintal)' },
  hovermode: 'closest'
};
return (
  <div className="bg-white rounded-lg shadow-md p-4">
    <Plot
      data={chartData}
      layout={layout}
      style={{ width: '100%', height: '400px' }}/>
```

```
    </div>
  );
};
export default PriceTrendChart;
```

A.4 Backend API Example

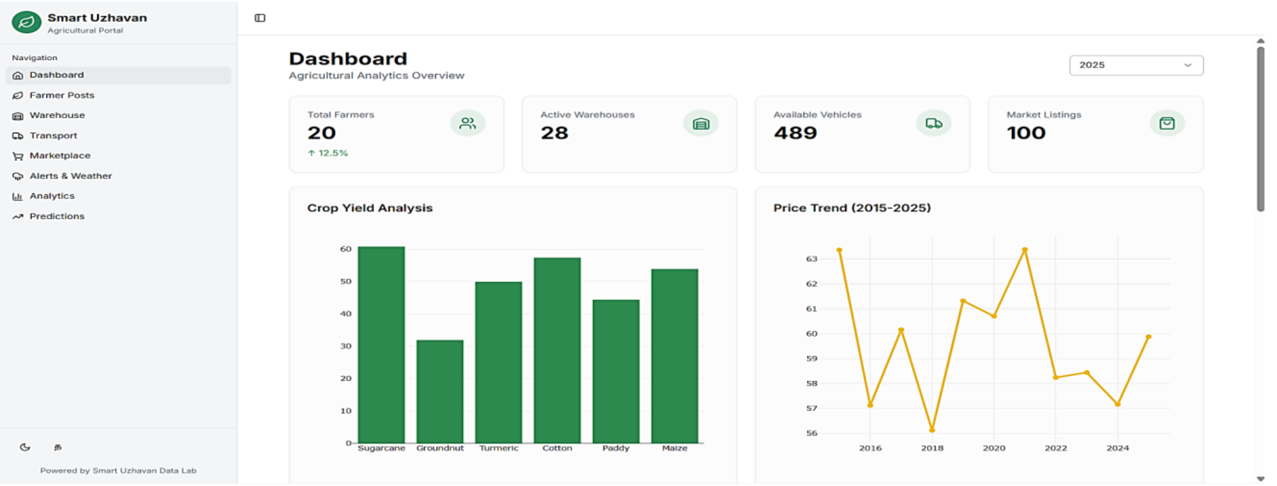
```
// cropRoutes.js
const express = require('express');
const router = express.Router();
const mysql = require('mysql2/promise');
const authMiddleware = require('../middleware/auth');
const pool = mysql.createPool({
  host: process.env.DB_HOST,
  user: process.env.DB_USER,
  password: process.env.DB_PASSWORD,
  database: process.env.DB_NAME,
  connectionLimit: 10
});
// POST new crop
router.post('/', authMiddleware, async (req, res) => {
  try {
    const { cropType, quantity, quality, harvestDate, location } = req.body;
    const userId = req.user.id;
    if (!cropType || !quantity || !harvestDate) {
      return res.status(400).json({ error: 'Missing required fields' });
    }
    const [farmers] = await pool.query(
      'SELECT FarmerID FROM Farmers WHERE UserID = ?',
      [userId]
    );
    if (farmers.length === 0) {
      return res.status(404).json({ error: 'Farmer profile not found' });
    const [result] = await pool.query(
```

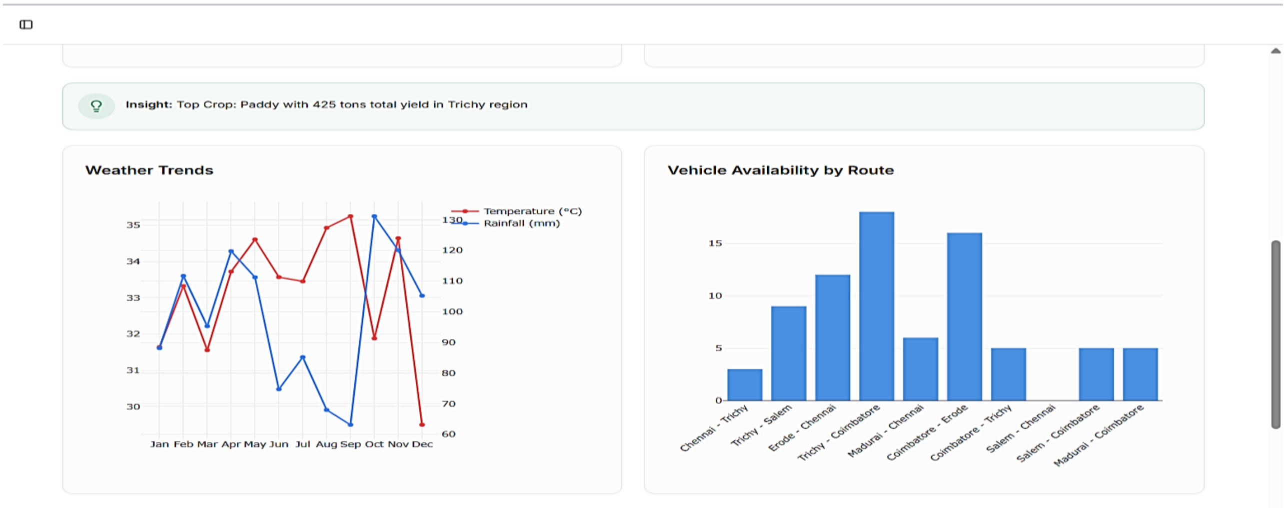
```
      `INSERT INTO Crops (FarmerID, CropType, Quantity, Quality,
       HarvestDate, Location, Status)
       VALUES (?, ?, ?, ?, ?, ?, 'Available')`,
      [farmers[0].FarmerID, cropType, quantity, quality, harvestDate, location]
    );
    res.status(201).json({
      message: 'Crop posted successfully',
      cropId: result.insertId
    });
  } catch (error) {
    console.error('Error creating crop:', error);
    res.status(500).json({ error: 'Failed to post crop' });
  }
});
module.exports = router;
```

APPENDIX B

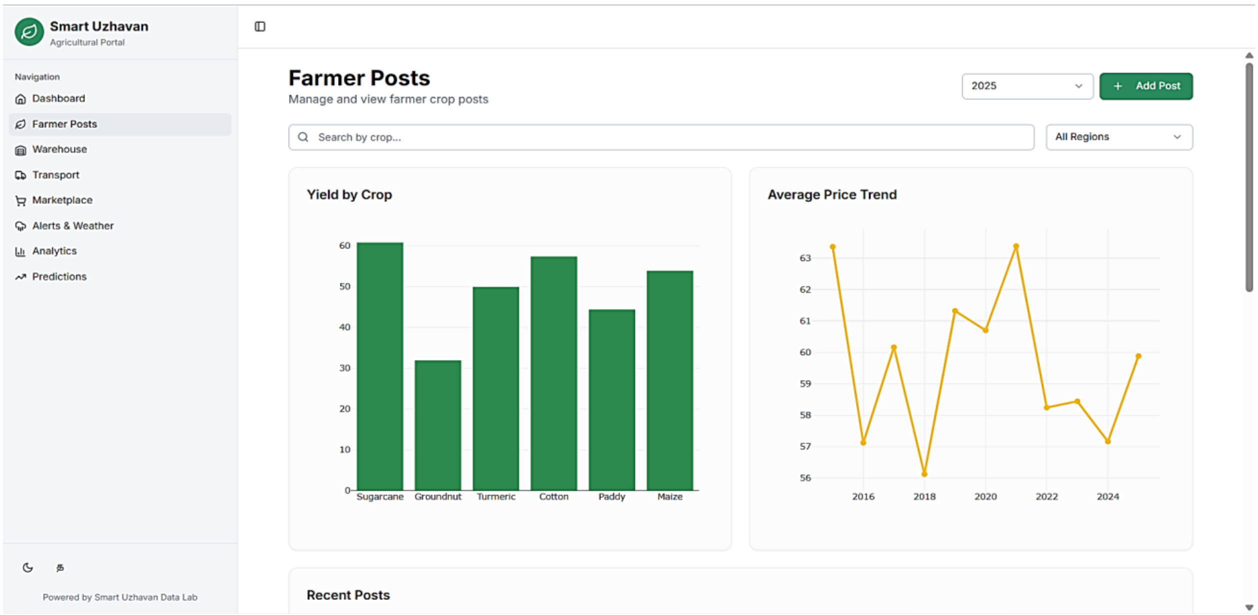
SCREENSHOTS

B.1 DASHBOARD





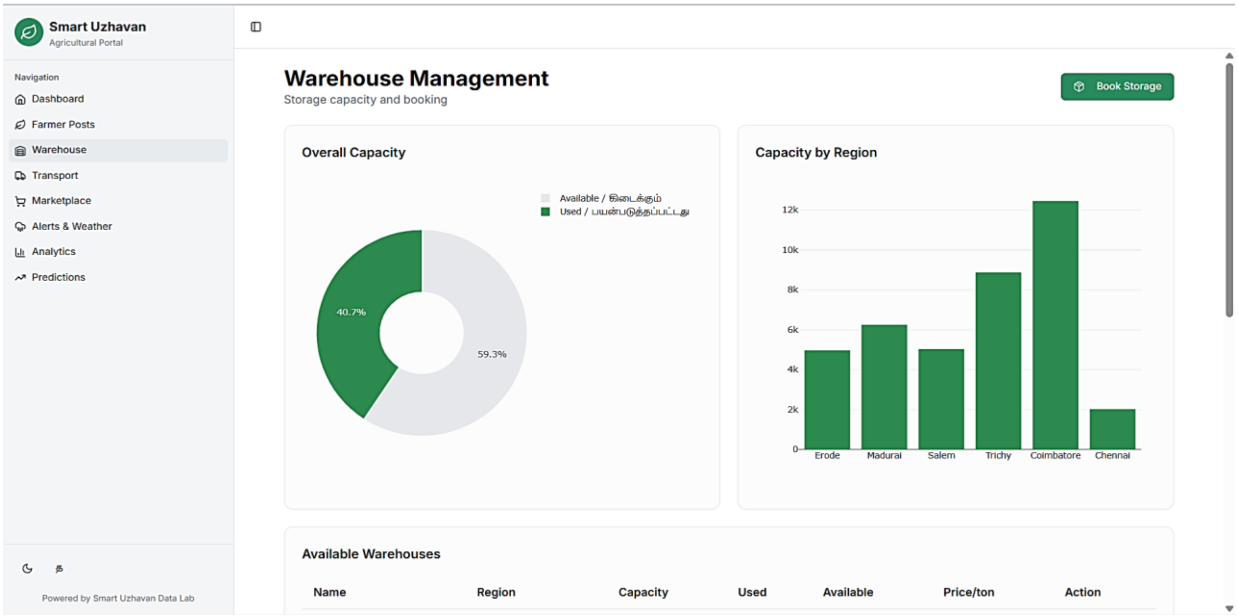
B.2 FARMER PAGE



Recent Posts

Crop	Region	Yield (kg)	Price (₹/kg)	Year
Sugarcane	Erode	6,385	₹43	2025
Groundnut	Chennai	2,825	₹64	2025
Groundnut	Madurai	4,928	₹64	2025
Turmeric	Coimbatore	5,273	₹73	2025
Groundnut	Trichy	6,553	₹26	2025
Groundnut	Erode	5,634	₹69	2025
Sugarcane	Coimbatore	8,739	₹66	2025
Cotton	Chennai	3,591	₹35	2025
Sugarcane	Coimbatore	7,253	₹84	2025
Paddy	Madurai	7,324	₹47	2025
Turmeric	Chennai	6,169	₹54	2025
Turmeric	Salem	3,267	₹88	2025
Maize	Madurai	5,503	₹40	2025
Turmeric	Salem	6,797	₹38	2025
Maize	Trichy	3,186	₹94	2025
Maize	Madurai	8,273	₹31	2025
Turmeric	Trichy	9,463	₹27	2025
Cotton	Chennai	3,519	₹38	2025
Groundnut	Madurai	2,010	₹96	2025
Cotton	Coimbatore	4,543	₹85	2025

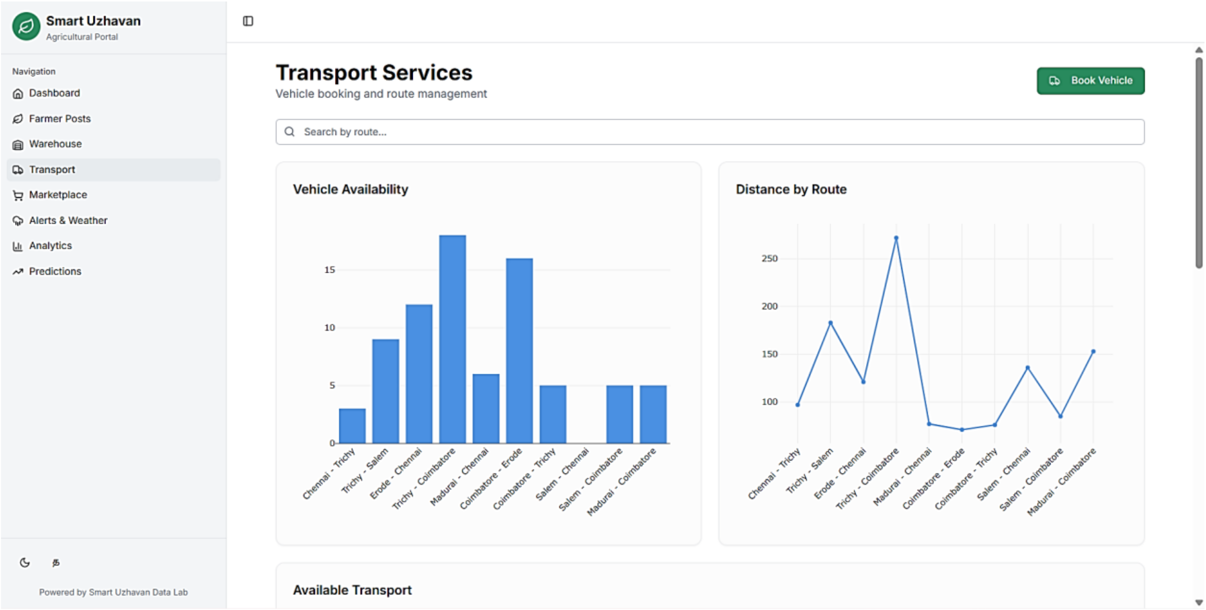
B.3 WAREHOUSE PAGE



Available Warehouses

Name	Region	Capacity	Used	Available	Price/ton	Action
Warehouse 1	Erode	1326t	549t	777t	₹1782	Book
Warehouse 2	Madurai	1363t	614t	749t	₹1594	Book
Warehouse 3	Erode	1595t	956t	639t	₹1052	Book
Warehouse 4	Salem	898t	48t	850t	₹939	Book
Warehouse 5	Trichy	1401t	349t	1052t	₹1212	Book
Warehouse 6	Madurai	1261t	556t	705t	₹1997	Book
Warehouse 7	Coimbatore	677t	175t	502t	₹1072	Book
Warehouse 8	Erode	1067t	315t	752t	₹1163	Book
Warehouse 9	Trichy	1748t	226t	1522t	₹537	Book
Warehouse 10	Coimbatore	1639t	273t	1366t	₹1995	Book
Warehouse 11	Chennai	1036t	191t	845t	₹807	Book
Warehouse 12	Coimbatore	1700t	140t	1560t	₹1505	Book
Warehouse 13	Coimbatore	1740t	573t	1167t	₹763	Book
Warehouse 14	Salem	673t	973t	-300t	₹1389	Book
Warehouse 15	Trichy	1544t	286t	1258t	₹769	Book

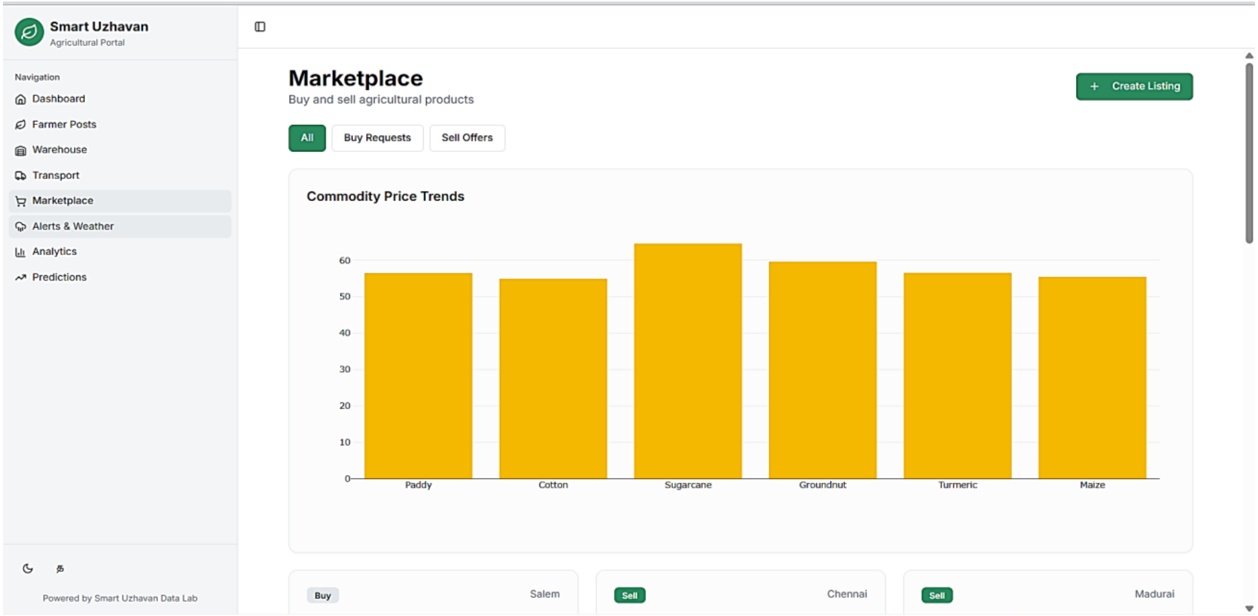
B.4 TRANSPORT PAGE

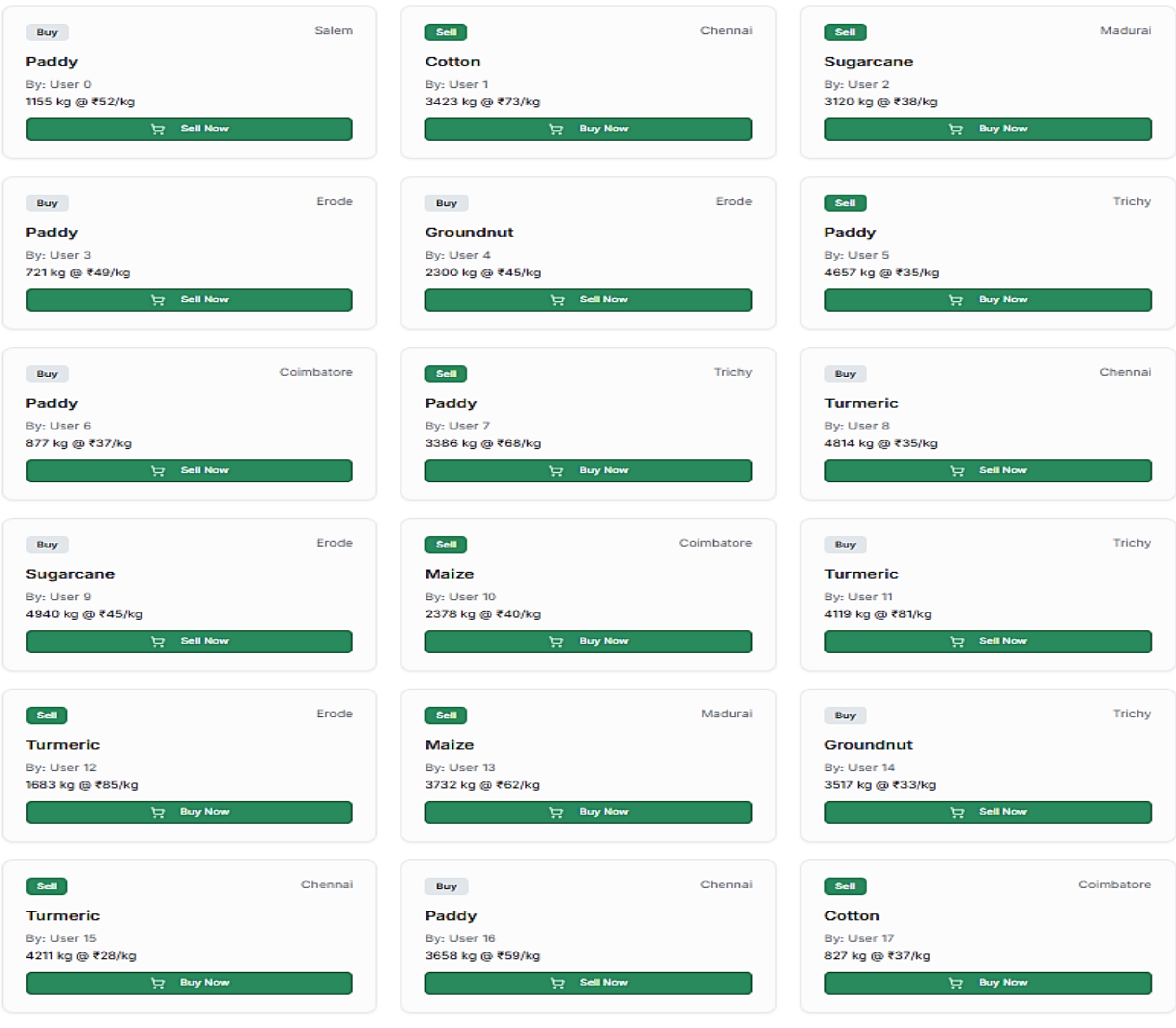


Available Transport

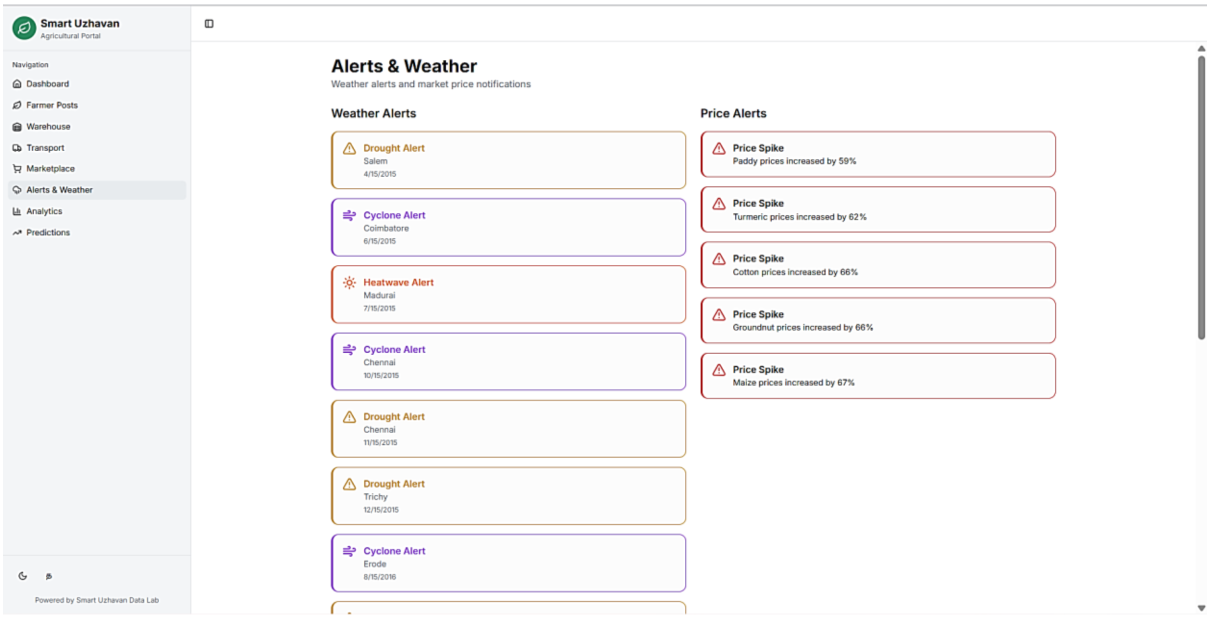
Route	Distance	Vehicle Type	Capacity	Available	Price/km	Action
Chennai - Trichy	97 km	Mini Truck	2t	3	₹15	Book
Trichy - Salem	183 km	Mini Truck	5t	9	₹25	Book
Erode - Chennai	121 km	Mini Truck	4t	12	₹24	Book
Trichy - Coimbatore	272 km	Trailer	8t	18	₹10	Book
Madurai - Chennai	77 km	Mini Truck	6t	6	₹10	Book
Coimbatore - Erode	71 km	Truck	3t	16	₹20	Book
Coimbatore - Trichy	76 km	Mini Truck	8t	5	₹11	Book
Salem - Chennai	136 km	Mini Truck	3t	0	₹21	Book
Salem - Coimbatore	85 km	Trailer	4t	5	₹26	Book
Madurai - Coimbatore	153 km	Trailer	8t	5	₹16	Book
Salem - Chennai	158 km	Truck	8t	14	₹22	Book
Erode - Trichy	185 km	Mini Truck	5t	4	₹19	Book
Madurai - Salem	347 km	Truck	4t	16	₹22	Book
Trichy - Coimbatore	82 km	Truck	7t	10	₹26	Book
Salem - Erode	246 km	Trailer	3t	2	₹38	Book
Coimbatore - Madurai	196 km	Trailer	5t	5	₹29	Book
Trichy - Salem	234 km	Truck	9t	9	₹35	Book
Coimbatore - Erode	256 km	Truck	9t	15	₹14	Book
Erode - Coimbatore	137 km	Truck	8t	1	₹10	Book
Salem - Madurai	101 km	Truck	5t	4	₹39	Book

B.5 MARKETPLACE PAGE

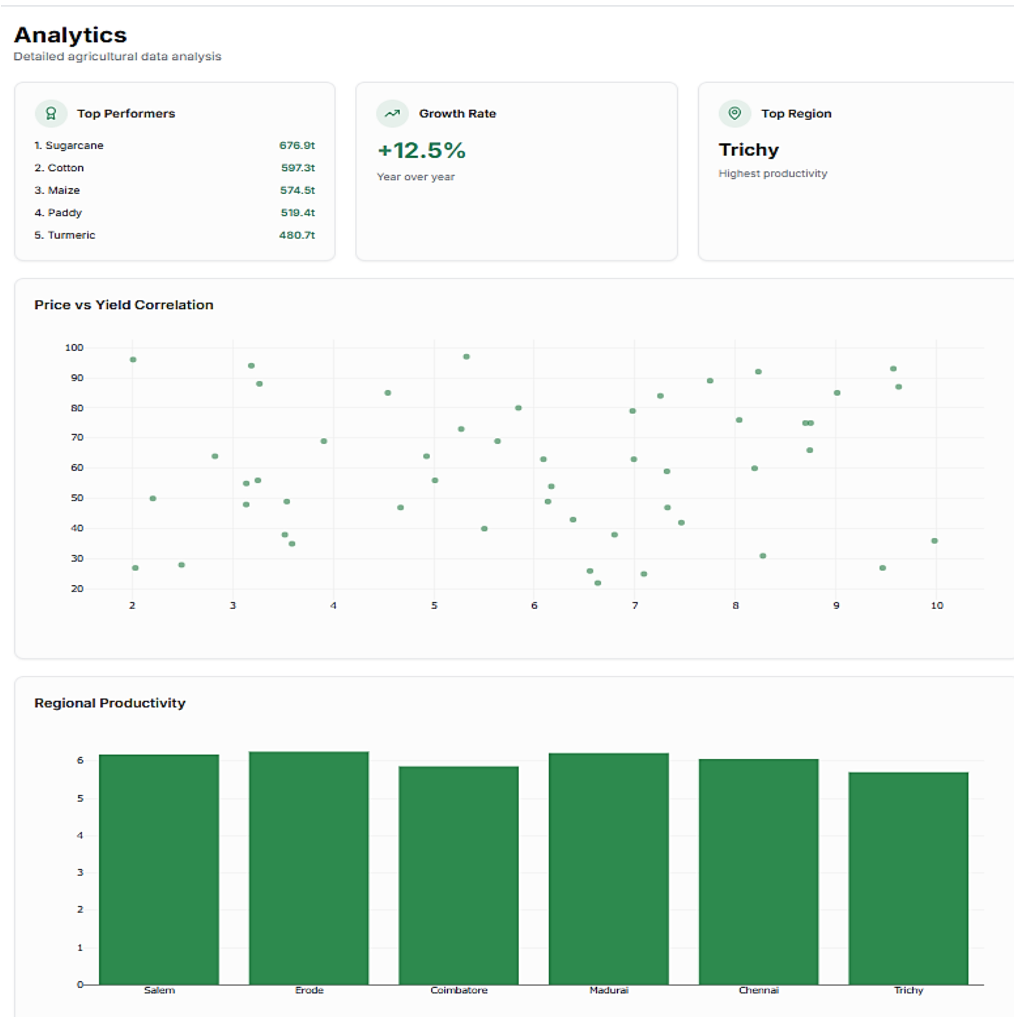




B.6 ALERTS PAGE



B.7 ANALYTICS PAGE



REFERENCES

1. **Aker, J. C., & Mbiti, I. M.** (2010). Mobile Phones and Economic Development in Africa. Journal of Economic Perspectives, 24(3), 207–232.
2. **Barbedo, J. G. A.** (2019). A Review on Use of UAVs and Imaging Sensors for Monitoring and Assessing Plant Stresses. Drones, 3(2), 40.
3. **Fountas, S., et al.** (2015). Farm Management Information Systems: Current Situation and Future Perspectives. Computers and Electronics in Agriculture, 115, 40–50.
4. **Kameswari, V. L., et al.** (2011). Rule Based Expert System for Agricultural Information System. International Journal of Advanced Computer Science and Applications (IJACSA), 2(3), 78–83.
5. **Khaki, S., & Wang, L.** (2019). Crop Yield Prediction Using Deep Neural Networks. Frontiers in Plant Science, 10, 621.
6. **Kumar, A. K., et al.** (2018). Mobile Applications in Agricultural Marketing. International Journal of Engineering and Technology, 7(3.12), 464–467.
7. **Liakos, K. G., et al.** (2018). Machine Learning in Agriculture: A Review. Sensors, 18(8), 2674.
8. **Mishra, S., et al.** (2020). Performance Evaluation of Machine Learning Model for Chronic Disease Datasets. Applied Sciences, 10(22), 8137.
9. **Mittal, S., & Mehar, M. S.** (2016). Socio-economic Factors Affecting Adoption of Modern ICT by Farmers in India. The Journal of Agricultural Education and Extension, 22(2), 199–212.
10. **Wolfert, S., et al.** (2017). Big Data in Smart Farming – A Review. Agricultural Systems, 153, 69–80.